NGUYEN NGOC TAM (NGOC TAM)

Senior Full-Stack Engineer & Real-Time Distributed Systems Architect (PHP / Laravel / Node.js / React / WebRTC / Cloud)

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EXECUTIVE ARCHITECTURAL SUMMARY

Senior-caliber Full-Stack Software Engineer & Real-Time Communication Systems Architect with deep production mastery in building high-concurrency Omnichannel Contact Centers, Banking-grade WebRTC Video/Voice infrastructure, and Enterprise CRM ecosystems. Proven track record across Tier-1 Banking (PVcomBank), Global Insurance (Shinhan Life), and Telecom Cloud (South Telecom / Worldfone4X). Recognized for reducing SonarQube code smells by 85%, eliminating critical XSS security vulnerabilities, cutting AWS cloud costs by 42%, and sustaining 99.99% production uptime across 100K+ daily real-time messaging and signaling events.

99.99% SLA Uptime	-85% Code Smells	<35ms P99 Latency	-42% Cloud FinOps Cost
Tier-1 Banking & Telecom	SonarQube & OWASP Hardened	WebRTC / WebSockets / Redis	AWS EC2, S3, Docker & K8s

CORE TECHNICAL ARCHITECTURE MATRIX

- **Languages & Runtimes:** PHP (7.x / 8.x Strict Types, JIT), TypeScript, JavaScript (ES2024+), Node.js, SQL, Bash.
- **Frameworks & UI Engineering:** Laravel, CodeIgniter, React, Express.js, Kendo UI, Tailwind CSS, RESTful APIs, MVVM & Clean Architecture.
- **Real-Time & Media Streaming:** WebRTC, Janus WebRTC Gateway, Jitsi Meet Mesh, WebSockets, SIP/VoIP Signaling, OTT APIs (Zalo, WhatsApp, LINE, LiveChat).
- **Databases, Queues & Caching:** PostgreSQL, MongoDB (Aggregation & Time-Series Schemas), SQL Server, Redis (Pub/Sub, Streams, Queues), Beanstalkd.
- **Cloud, DevSecOps & Quality:** AWS (EC2, S3), Docker, Kubernetes (K8s), GitLab CI/CD, GitHub Actions, Nginx, Linux, SonarQube, PHPUnit, JWT/OAuth2, OWASP XSS/CSRF Mitigation.

PROFESSIONAL LEADERSHIP & WORK EXPERIENCE

SOUTH TELECOM — *Senior Full-Stack Developer & Real-Time Systems Lead / Scrum Facilitator* (07/2022 – Present)

- Architected and scaled core modules for enterprise omnichannel contact center platforms serving major banking, finance, and telecom enterprises.
- Engineered real-time bi-directional communication pipelines integrating Voice, WebRTC Video, Zalo OA, WhatsApp Business, LINE, and LiveChat with <35ms P99 routing latency.
- Spearheaded Cloud FinOps initiatives across AWS (EC2, S3) and containerized workloads, optimizing resource allocation and reducing monthly cloud expenditure by ~42%.
- Designed fault-tolerant webhook & bi-directional CRM synchronization engines for Salesforce and HubSpot using Redis queues and Beanstalkd workers.
- Served as Scrum Host (Agile Facilitator) & Technical Mentor: led sprint ceremonies, unblocked cross-functional bottlenecks across Product/QA/DevOps, and onboarded new engineers to production readiness.

R-DIGITAL — *Backend Software Engineer (Associate / Intern)* (04/2022 – 07/2022)

- Collaborated with cross-functional frontend and backend squads to design modular API contracts, optimize relational queries, and establish daily engineering delivery metrics.

FLAGSHIP ENTERPRISE PRODUCTION CASE STUDIES

1. Worldfone4X — Enterprise Omnichannel Contact Center Ecosystem (Team: 10 | Stack: PHP 8, MongoDB, Redis, Beanstalkd, WebSockets, Kendo UI)

- Unified voice, social messaging (Zalo, WhatsApp, LINE, LiveChat), and Salesforce/HubSpot CRM workflows into a single high-availability platform. Optimized Omnichat message dispatch queues to handle peak concurrency bursts without message loss.

2. PVcomBank — Banking-Grade WebRTC Video eKYC & Encrypted Communication Platform (Team: 3 | Stack: PHP, WebRTC, Custom Signaling, MongoDB, Redis, WebSockets)

- Built a standalone banking communication & secure video call suite for PVcomBank. Engineered custom WebRTC signaling, encrypted media transport, load-balanced WebSocket clusters, and strict token-based session verification integrated with core banking APIs.

3. Shinhan Life — Financial Enterprise Customization & DevSecOps Hardening (Team: 2 | Stack: PHP, SonarQube, PHPUnit, MongoDB, Redis, WebSockets)

- Led onsite deployment and security audit remediation for Shinhan Life. Eliminated critical XSS vulnerabilities, reduced SonarQube code smells by 85%, implemented comprehensive PHPUnit test coverage for mission-critical insurance workflows, and synchronized internal enterprise APIs.

4. Distributed Video Room Integration System (Janus WebRTC + Jitsi Mesh) (Team: 2 | Stack: Laravel, Janus WebRTC, Jitsi, Redis Queues, MongoDB, WebSockets)

- Architected an asynchronous real-time video room event bridge using Laravel Queues (Redis) to process high-frequency Janus WebRTC server events with zero downtime. Normalized occupant metadata between Janus and Jitsi and automated stale session purging via Laravel Task Scheduling.

EDUCATION

Industrial University of Ho Chi Minh City (IUH) — Bachelor of Information Technology (2019 – 2022) | Ho Chi Minh City, Vietnam